GCET
G H PATEL COLLEGE OF ENGINEERING & TECHNOLOGY

MINI PROJECT - 202040601

CVM UNIVERSITY

MedInventory – Medical Inventory System

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Introduction

➢ *Problem Statement*

- In today's healthcare facilities and pharmacies, managing medical inventory efficiently remains a critical challenge that directly impacts patient care and business operations. Traditional manual methods of tracking medical supplies, monitoring expiration dates, and maintaining optimal stock levels are prone to errors, time-consuming, and often result in significant financial losses. Healthcare providers struggle with stockouts of essential medicines, expired inventory, and inefficient resource allocation, which can compromise patient care quality and safety.

- Currently, staff members spend considerable time on manual inventory counts, paperwork, and tracking expiration dates without real-time visibility into stock levels. This leads to emergency purchases at premium prices, wastage due to expired medications, and potential risks of distributing expired products to patients. The lack of automated alerts for low stock and near-expiry items forces reactive rather than proactive inventory management.

Introduction

➢ *Problem Objective*

- **Inventory Management:** Create a real-time medical inventory tracking system with automated alerts for low stock and expiring items.
- **Digital Transformation:** Digitize manual inventory processes to reduce errors, save time, and improve efficiency in healthcare facilities.
- **Quality Control:** Ensure patient safety by monitoring expiration dates and maintaining proper stock levels of medical supplies.
- **Data Analytics:** Provide insights through dashboards, reports, and analytics for better inventory decision-making.
- **User Experience**: Deliver an intuitive interface for healthcare staff to manage inventory with minimal training**.**

Literature Review

➢ Related Work

1. **Traditional Systems**
   - Manual record keeping
   - Paper-based inventory tracking
   - High error rates and inefficiencies
   - Limited real-time visibility
2. **Digital Transformation**
   - Computerized systems (1990s-2000s)
   - Early database management systems
   - Basic stock tracking software
   - Limited automation capabilities
3. **Modern Stack Implementation**
   - MERN Stack Architecture
   - Real-time data synchronization
   - Cloud database integration
   - Responsive design principles

Literature Review

➢ Related Work

4. Automation

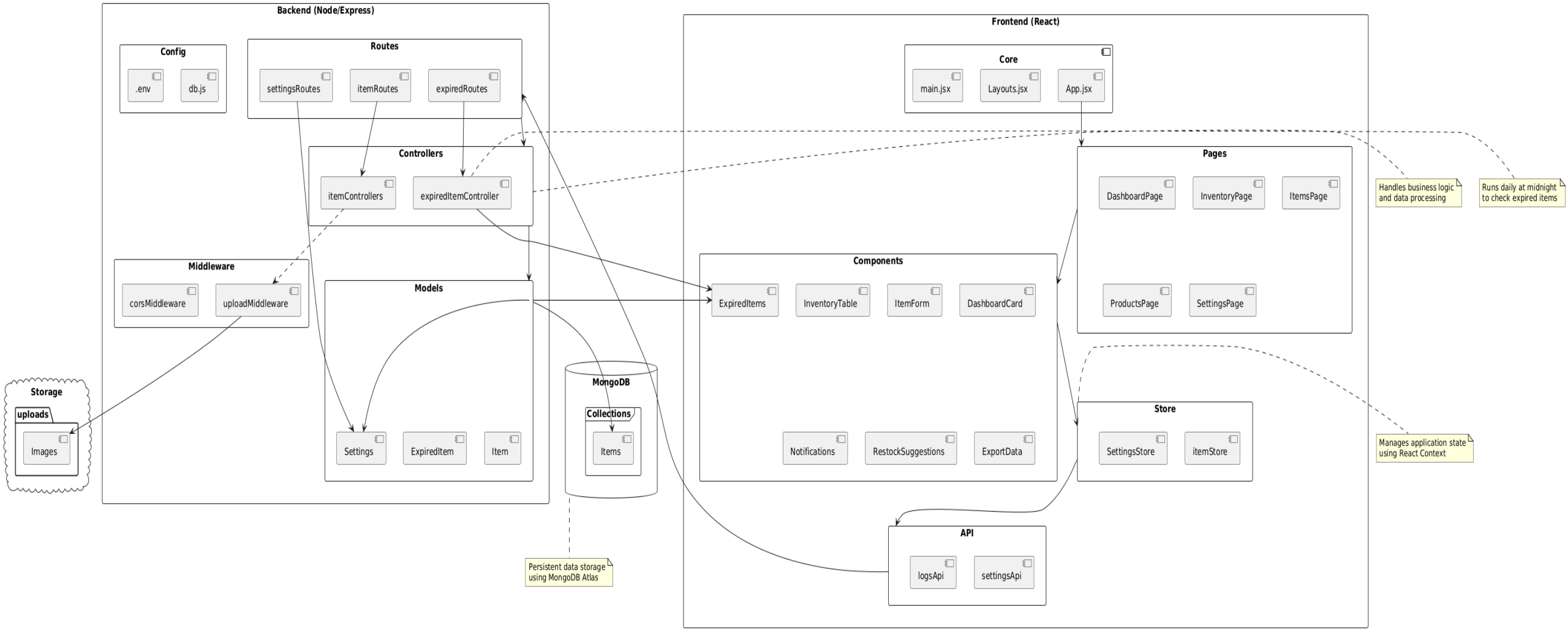
- Automated expiry tracking
- Real-time notifications

5. Data Management

- Advanced filtering
- Export capabilities

System Design

Architecture Diagram



System Design

➢ Model Overview

1. **Item Model** : Manages active inventory items
2. **ExpiredItem Model :** Tracks expired inventory items
3. **Settings Model** : Manages application settings and preference

Key Relationships:

- **Item → ExpiredItem** :
  - Items automatically move to ExpiredItem when expired

- **Item → Settings** :
  - Settings control item thresholds
  - Notifications based on settings

Implementation

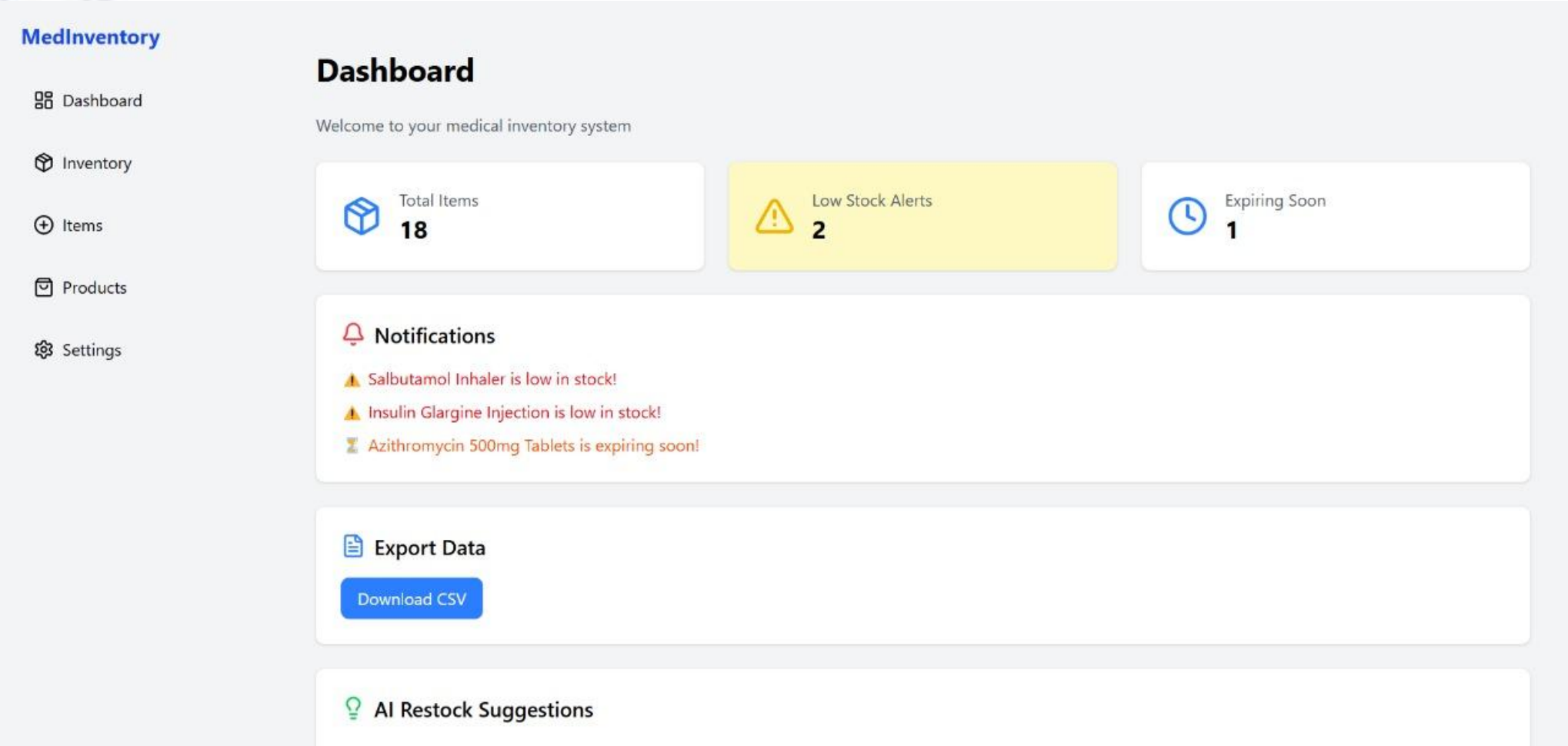
➢ Technologies Used

The **MedInventry** is implemented using the following technologies and tools:

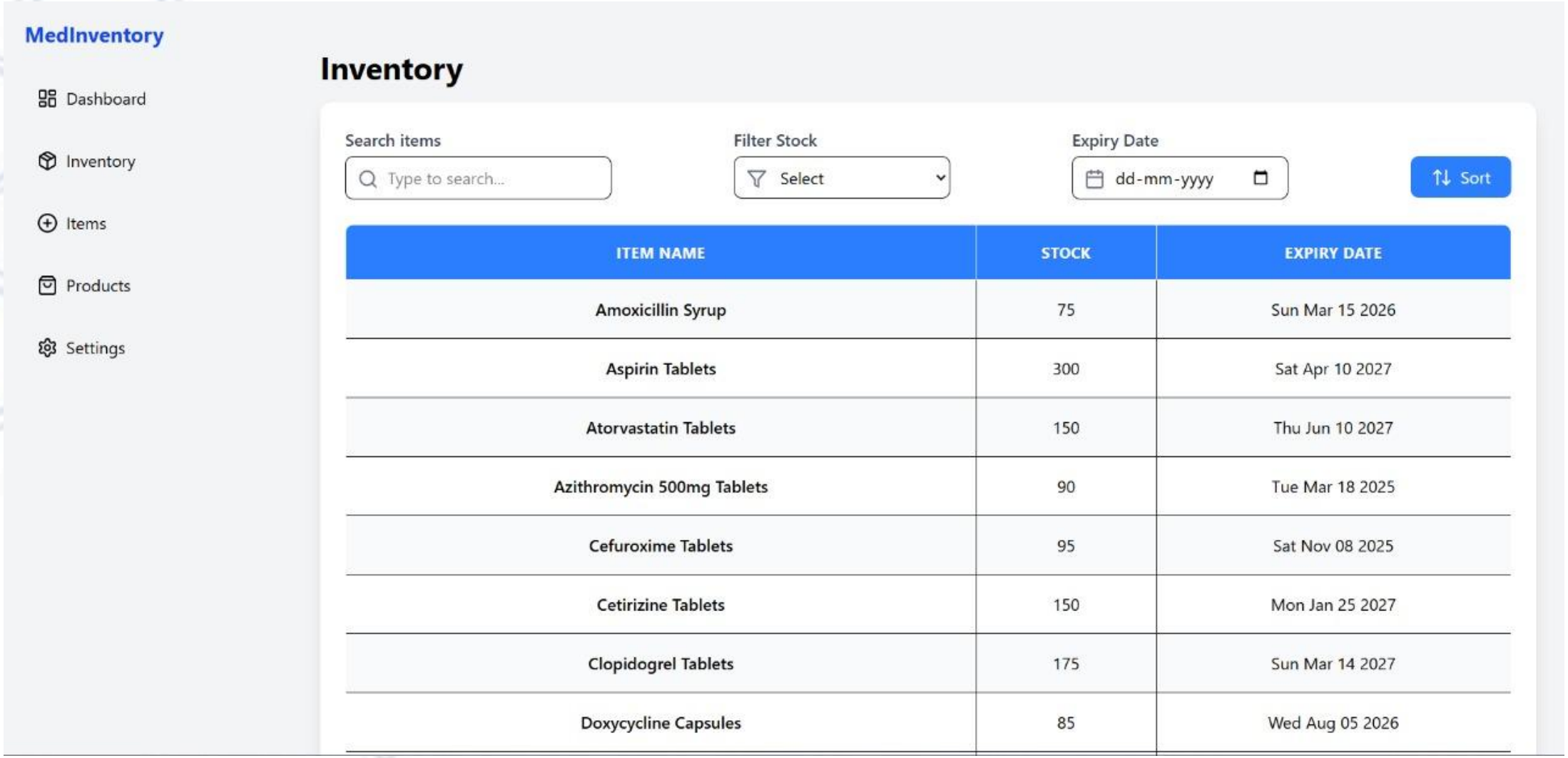
- **MERN Stack** –
  - React.js (v19.0.0)
  - Express.js (v4.21.2)
  - MongoDB (Mongoose v8.10.0)
  - Node.js
- **Additional Frontend Libraries** –
  - Lucide React – Icons
  - React Hot Toast – Notifications
  - TailwindCSS (v4.0.4)
- **Additional Backend Packages** –
  - Multer - File upload handling
  - CORS - Cross-origin resource sharing
  - Dotenv - Environment variables

Implementation

- **ScreenShots**

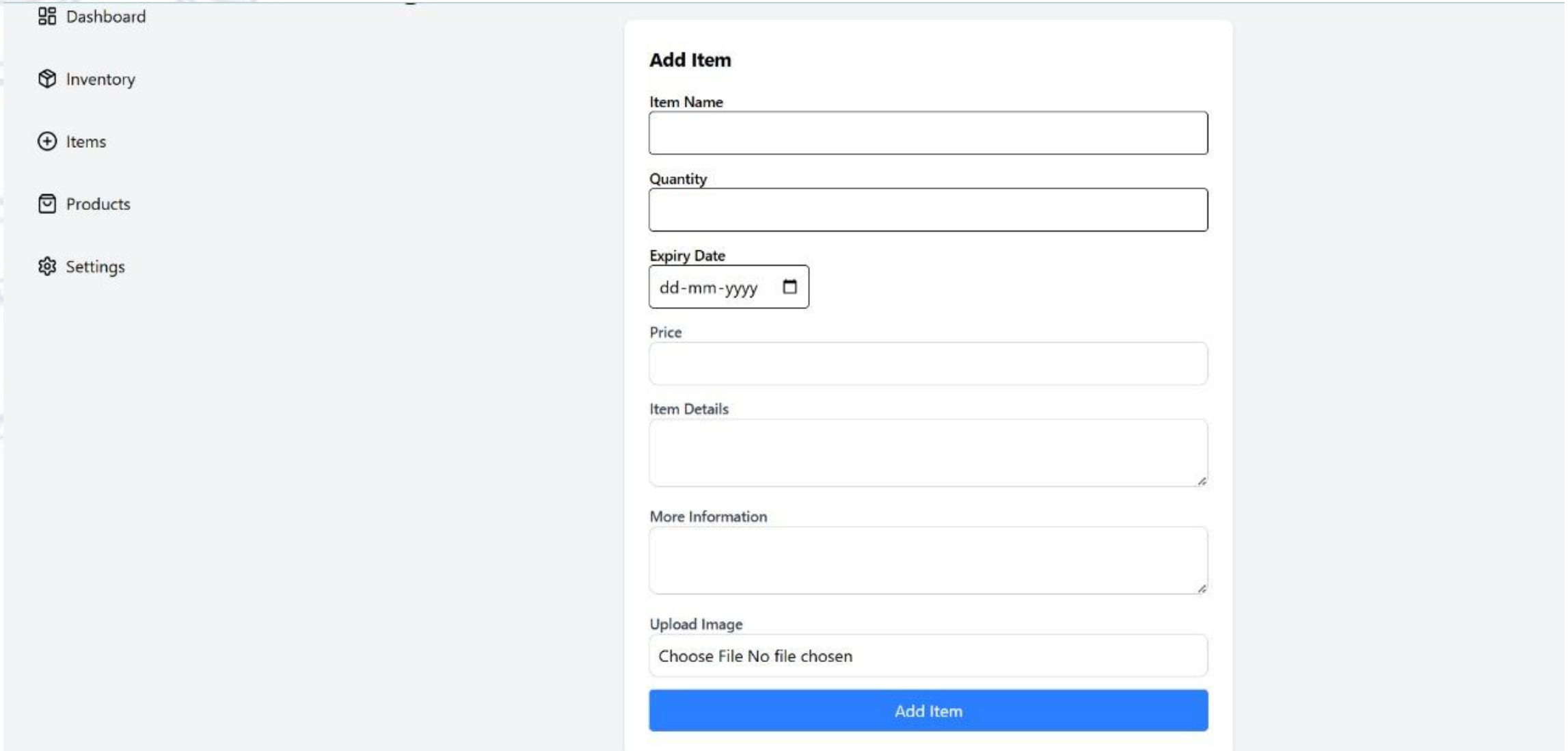


Implementation



ITEM NAME	STOCK	EXPIRY DATE
Amoxicillin Syrup	75	Sun Mar 15 2026
Aspirin Tablets	300	Sat Apr 10 2027
Atorvastatin Tablets	150	Thu Jun 10 2027
Azithromycin 500mg Tablets	90	Tue Mar 18 2025
Cefuroxime Tablets	95	Sat Nov 08 2025
Cetirizine Tablets	150	Mon Jan 25 2027
Clopidogrel Tablets	175	Sun Mar 14 2027
Doxycycline Capsules	85	Wed Aug 05 2026

Implementation



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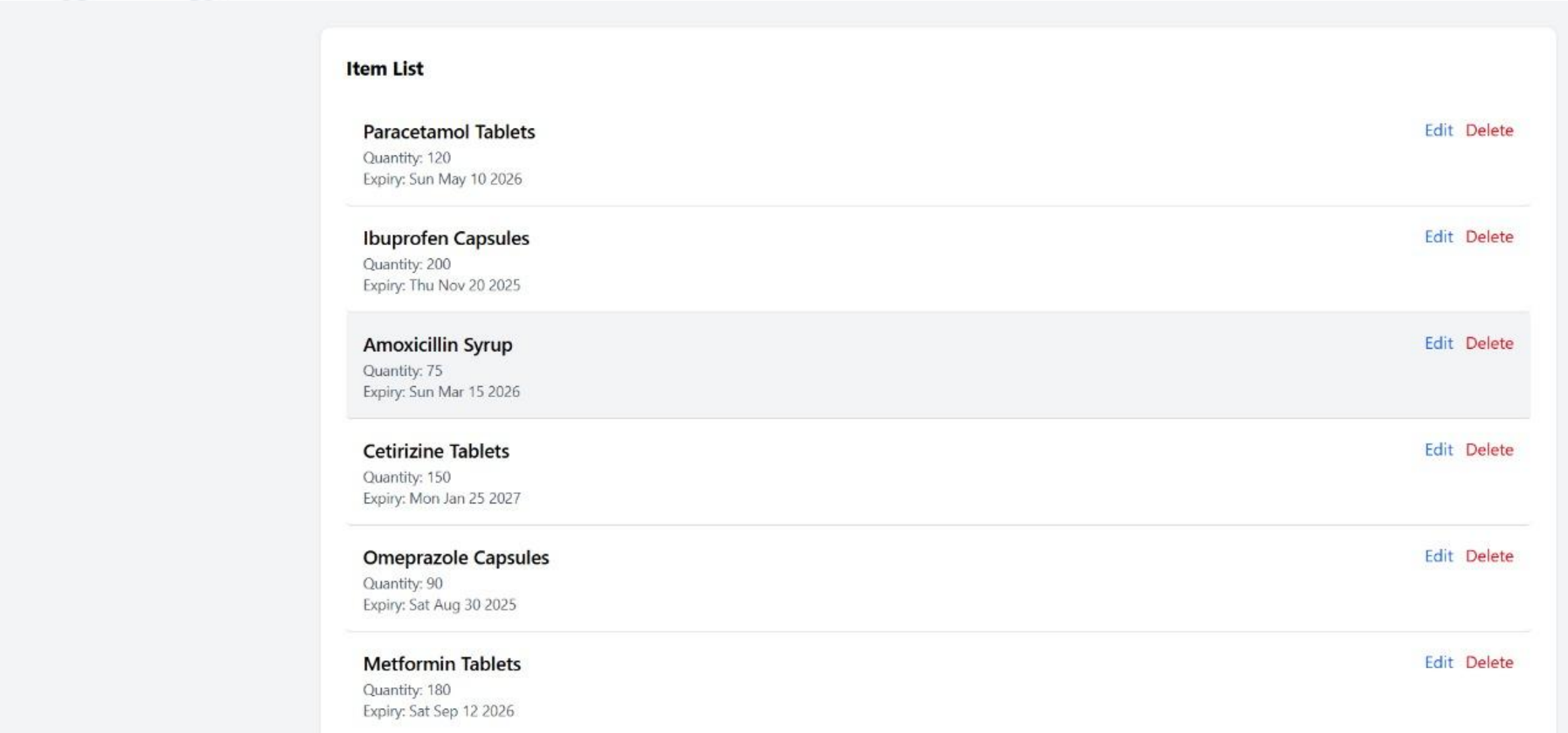

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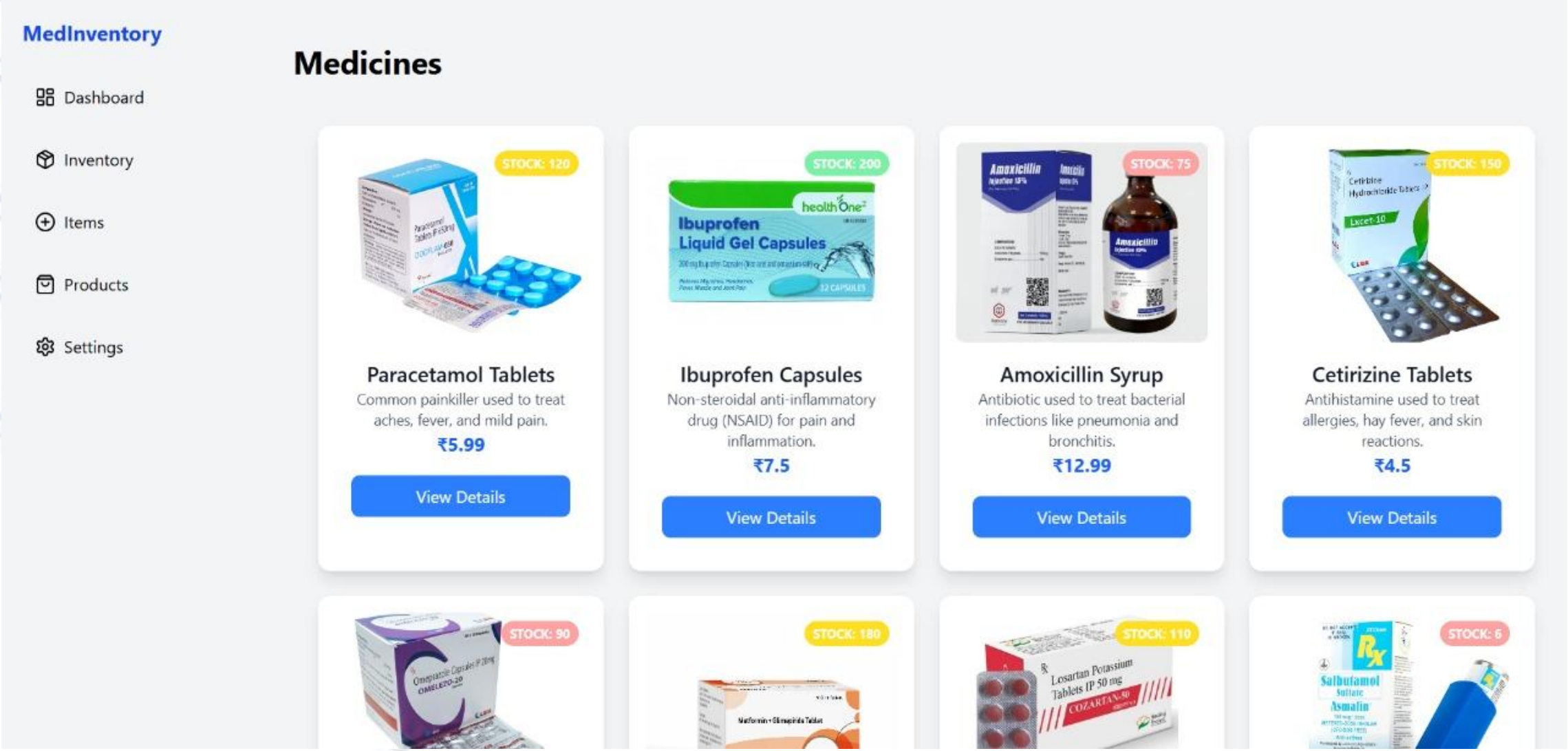

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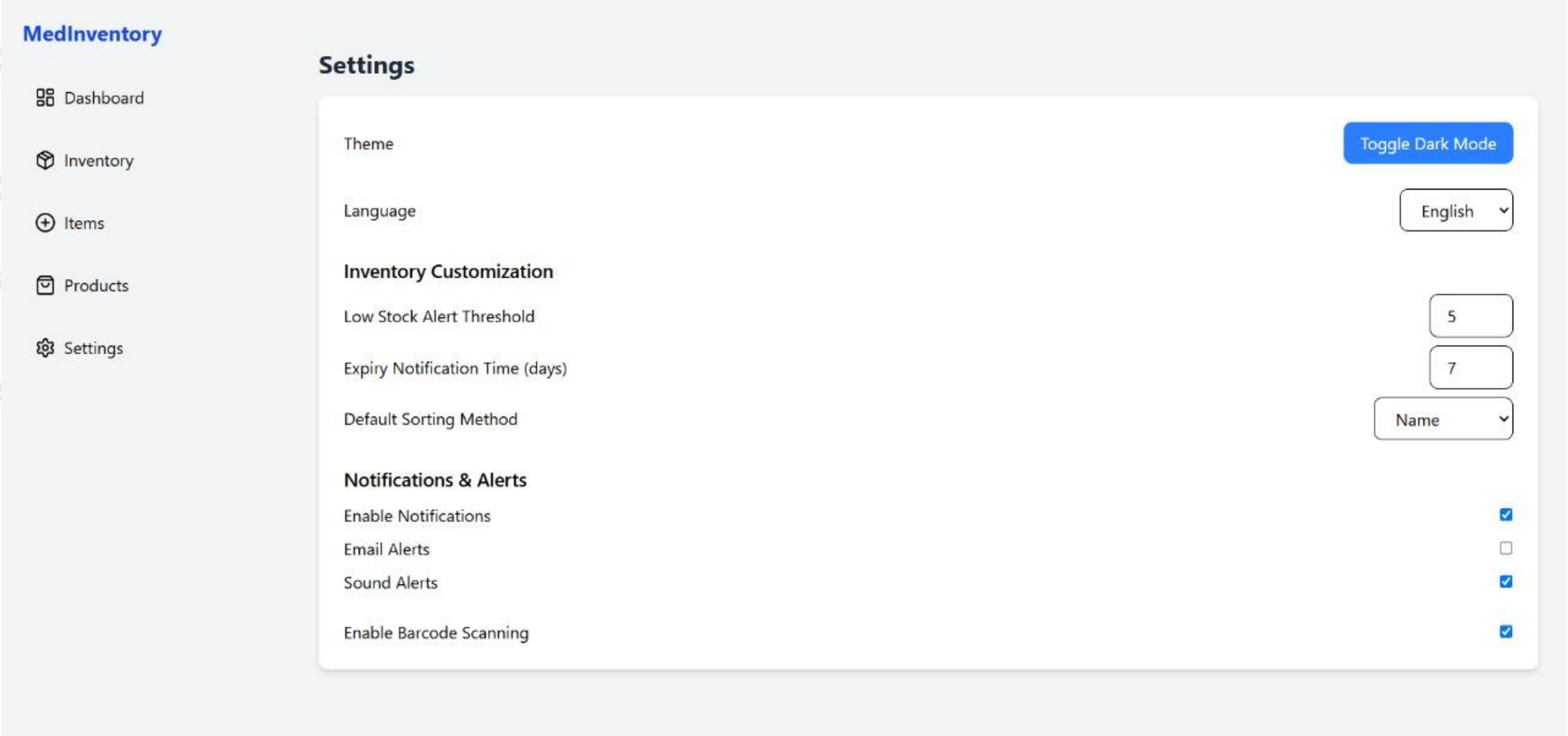

Conclusion and Future Scope

Conclusion:

The Medical Inventory Management System provides an efficient and user-friendly platform for managing medical supplies, reducing wastage, and ensuring seamless inventory tracking. With its implementation, healthcare institutions can streamline their inventory processes, minimize errors, and improve resource utilization. The system integrates modern web technologies, ensuring a scalable and responsive user experience. By utilizing a database-driven approach, it offers real-time updates and inventory insights, enhancing overall operational efficiency.

Future Scope:

- AI-Powered Demand Forecasting
- Supplier and Order Management
- Automated Alerts and Reports
- Adding Medicine recommend system

Final Thoughts:

The development of this system marks a significant step toward digital transformation in medical inventory management. It not only simplifies stock tracking but also ensures optimal resource utilization, reducing inefficiencies and improving patient care. As technology evolves, continuous improvements and innovations will further enhance its capabilities, making it a vital tool for modern healthcare facilities.

References

➢ Research Papers & Journals

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Q&A